

Project Planning & Lifecycle Roadmap

Project: **Credit Card Fraud Detection System** • Developer: **Bhargavasai**

PRIMARY OBJECTIVE GOAL

Design, build, optimize, and deploy an end-to-end operational Machine Learning web application pipeline engineered specifically to classify real-time financial transaction streams into legitimate or fraudulent vectors.

01

Dataset Collection & Aggregation

Phase: Data

Sourcing highly granular transactional records containing time matrices, numeric value properties, and anonymized credit dimensions.

02

Data Preprocessing & Data Cleaning

Phase: Prep

Executing full integrity passes including handling missing data items, removing duplicate instances, and scaling metrics uniformly via StandardScaler.

03

Exploratory Data Analysis (EDA)

Phase: Analysis

Analyzing hidden feature structures and class imbalances through distribution histograms, scatter charts, and multi-variable correlation heatmaps.

04

Machine Learning Model Building

Phase: Modeling

Training robust multi-algorithmic classification boundaries (Logistic Regression, Decision Trees, Random Forests, and XGBoost structures).

05

Flask Application Development

Phase: Engine

Constructing responsive HTML/CSS presentation wrappers and writing Flask API routing modules to load serialized pipelines via Joblib.

06

System Testing & Pipeline Evaluation

Phase: Audit

Auditing computational behaviors via classification reports, error logs, accuracy indicators, and confusion matrix testing arrays.

07

Production Cloud Deployment

Phase: DevOps

Pushing version-controlled code frameworks onto GitHub repositories and establishing active live hosting URLs via the Render cloud environment.

